Network, Signals & Systems (EC5.101)

- Faculty
  - Santosh Nannuru (santosh.nannuru@iiit.ac.in)
  - Anshu Sarje (anshu.sarje@iiit.ac.in)
- 4 credits: (3-1-0-4) – Lectures (3 hrs.) + Tutorials (1 hr.)
- Lectures: Tue, Thu, Sat (11:45 – 12:40)
- Tutorials: Tue (2 – 3 pm)
- Platform: Moodle for course material/lecture notes/assignments

Team

- Faculty
  - Santosh Nannuru [Signals & Systems]
    - Office Hrs: Thu 10:30 – 11:30 (Vindhya, A3-205)
    - https://sites.google.com/view/santoshnannuru/home
  - Anshu Sarje [Networks]
    - https://sites.google.com/view/iiit-hyd-cvest-nanodevice/home
- Teaching Assistants (TAs)
  - Ram Gopal (jalluri.gopal@students.iiit.ac.in, Office Hrs: Sat 4 – 5, workspace opposite to library)
  - Mallika Garg (mallika.garg@students.iiit.ac.in, Office Hrs: Sat 4 – 5, Left workspace)
  - Shirisha Reddy (shirisha.v@students.iiit.ac.in, Office Hrs: Sat 3 – 4, A5 321 CVEST)
  - Nobhendu Chowdhury (nobhendu.chowdhury@research.iiit.ac.in, Sat 3 – 4, A3-218 SPCRC)
  - Meghana Tedla (meghana.tedla@students.iiit.ac.in, Wed 4:30 – 5:30, SERC, T-HUB 4th floor)

Textbooks

- Primary Textbooks

1) Engineering Circuit Analysis (ECA)
    - 8th Edition, Tata McGraw-Hill
    - By W.H. Hyatt, J.E. Kimmerley, and S.M. Durbin

2) Signals and Systems (SAS)
    - 2nd Edition, Prentice Hall
    - By Oppenheim, Willsky and Nawab

Teaching style

- Lectures
  - Instructor will use mix of slides and board
  - Students strongly encouraged to take notes & solve practice problems
  - A short question will be posted at beginning of class
- Tutorials
  - Problem solving sessions (conducted by TAs)
- Schedule
  - ~ 6 classes (Signals, Santosh)
  - ~ 12 classes (Signals & Systems, Santosh)
  - ~ 18 classes (Networks, Anshu)

Evaluation

Evaluation type	Weightage (%)
Quiz-1	15*
Mid exam	25*
Quiz-2	15*
Final exam	25*
Assignments / Short Projects	20

Grading will be relative with absolute cut-off for A grade (90+ Marks)

Assignments

- Weekly assignments
- Typically due in a week
- 10% penalty/day for late submission (up to 3 days)

Prerequisites

- Complex numbers
- Basic calculus
- Vectors and matrices
- Solving system of equations

Syllabus

Signals, representation, sinusoids, and Fourier series
Circuit elements, Network theorems
Transient and Steady state analysis
Sinusoidal input and phasors
Two port network
Systems and representations
Convolution integral and impulse response
Transfer function – Laplace transform, poles and zeros

After completion of this course successfully, the students will be able to …

1. Describe various circuit elements (R, C, L), supply (current, voltage), devices (op amp, diode)
2. Explain the operation and characteristics of each circuit element, behavior in specific circuit configuration (DC, AC, series, parallel, mixed)
3. Calculate equivalent circuit parameters (Thevenin, Norton), node voltages, branch currents etc. using reduction, KCL, KVL and reduction techniques
4. Calculate circuit response (steady state, transient) to various input stimulation. Calculate and understand the concept of time constant for RC, RL and RLC circuits
5. Demonstrate understanding of and calculate Power, Energy, Loss and phasors w.r.t. circuit
6. Describe signals using various representations including Fourier series representation for periodic signals
7. Describe systems abstractly using block diagrams and differential equations
8. Apply convolution operation and impulse responses for system analysis
9. Analyze signals and systems using Laplace transform representation
10. Apply the above concepts to analyze and solve a real-life circuit/systems problem